# (3) When Controlled Experiments Are Not Possible 137

## (1) What Does the Section Cover?

Core Idea:

Controlled experiments (like A/B tests) are the gold standard for establishing causality. But in some cases, they're not possible — either due to ethical, technical, cost, or logistical constraints.

So, when you can't randomize, you turn to observational causal studies.

## (2) When Are Controlled Experiments NOT Possible?

Here are the main scenarios the authors point out:

### 1. Ethical Constraints

- You cannot assign users to a group where they might be harmed.
- Examples: You can't randomly give some users a buggy payment system to see if it reduces purchases — that's unethical.

### 2. Technical Limitations

- Your infrastructure can't support random assignment or tracking.
- Examples: An older legacy system that doesn't support user-level targeting or logging.

### 3. Low Sample Size

- If the user base is too small, running an experiment may take too long to get statistically significant results.

### 4. One-Time or Rare Events

- Some things happen only once, so you can't do randomized comparisons.
- Example: A one-time Super Bowl ad.

5. Cost-Prohibitive.

- Implementation two versions of a system just for an experiment may be too expensive or labor-intensive

# Real-World Analogy

> Imagine a doctor wants to know if smoking causes lung cancer. But they can't force people to smoke (unethical), and people already choose to smoke or not.

This is where observational causal studies step in — they observe natural behavior and apply statistical corrections to estimate causality, even without randomization.

(3) Core Mathematical Concept: Counterfactuals in Causal Inference. (= Causal effect)

# Real-world Analogy:

Imagine you are trying to figure out whether taking a coding bootcamp helps you get a job.

Let's say:

- You took the bootcamp and got a job.
- But how do we know you wouldn't have gotten a job anyway?

** That "what if I hadn't taken the bootcamp?" is a counterfactual — the outcome we didn't observe. **

# Notation

For a person (or unit) $i$, define:

- $Y_i(1)$: outcome if they receive the treatment (e.g., person takes the bootcamp)

$Y_i(0)$ : outcome if they don't receive the treatment (e.g. person doesn't take the bootcamp)

Then:

☑ Causal effect for person $i = Y_i(1) - Y_i(0)$

only one is observed per person.
The other is the counterfactual.

# Problem

We only observe one of these two for each person.

- if person did take the bootcamp, we observe $Y_i(1)$
- If they didn't, we observe $Y_i(0)$

* so the other one is missing - it's the counterfactual. *

## # Goal of Causal Inference

→ Causal inference is the science of answering:
"Did X cause Y?"

→ This is different from correlation, because correlation does not imply causation.

## # Average Treatment Effect (ATE)

. Estimating the effect on one person involves a counterfactual - and is usually impossible directly. But when we do it over a group, we can estimate average causal effects (like ATE) under assumptions — and that's what makes it causal inference.

The average effect across a population:

$$ATE = E[Y(1) - Y(0)]$$

we estimate ATE by comparing two groups

$$\widehat{ATE} = E[Y|T=1] - E[Y|T=0]$$

* But this only works if both group are comparable *

# So when is this comparison valid?

☑ when unconfoundedness holds:

$$(Y(1), Y(0)) \perp\!\!\!\perp T|X$$

Now we can estimate ATE within each X group, then averages across the population:

$$ATE = E_X[E[Y|T=1, X] - E[Y|T=0, X]]$$

$$= E_X[E[Y(1)|X] - E[Y(0)|X]]$$

This is called the back-door criterion in causal inference (Pearl).

Example:

Imagine

| Person | Bootcamp | Got Job? | Age | Degree |
|---|---|---|---|---|
| A | Yes | ✓ | 25 | Yes |
| B | No | X | 25 | Yes |

Here, B is like a minnor version of A — same age, same degree, but didn't take bootcamp

So we say:

| A's counterfactual is approximately what happened to B.

if you have enough data, you can match people who are:

- similar on all important characteristics.
- Differ only in whether they got the treatment (e.g. bootcamp).

(3) Linked chapter and topic

(1) Linked to chapter 12 concepts → Design for observational causal studies.

(2) A/B testing

↳ In randomized experiments, you rely on random assignment to balance all confounder.

↳ In observational studies, you must manually correct for confounder (e.g. age, gender, locations).

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## ② Design for Observational Causal studies:

This sections introduces how to design observational studies to estimate causal effects, especially when randomized experiments (A/B tests) aren't possible.

Core Goal:

> To mimic the effect of a randomized controlled experiment using real-world data where treatment assignment is not random.

Since we can't control who gets treatment, we must for the differences between groups in clever ways.

### 1. Matching

= Goal: Pair each treated unit with a similar untreated unit based on observed characteristics (age, income, etc)

# Core idea:

> Match each treated person with a similar untreated person based on characteristics (like age, income, prior behavior).
> Then, use the untreated person's outcome as the counterfactual for the treated one.

# Why do we do this?

We want to estimate:

$$\text{Effect} = Y_i(1) - Y_i(0)$$

But we never observe both for one person.
So for a person who got treated, we look for someone who didn't - and use their outcome to estimate what

would've happened without treatment.
That's matching.

# Real-world Example
You want to know if a COVID vaccine reduces death.
You can't observe what would've happened to a vaccinated person if they weren't vaccinated

# Math - Assumption

Matching works when:

$$Y_i(1), Y_i(0) \perp T_i \mid X_i$$

Meaning:

| once we control for X (age, gender, etc) treatment is as if random

so treated and untreated people with same X are comparable.

## (2) Stratification / Subclassification

Goal: Divide data into "bins" or strata based on variables age group, income bracket, etc. Then compare treated vs. untreated within each stratum.

# Real-world Analogy
You don't compare all treated vs. untreated directly
Instead, you compare:
- 20-30 years-olds treated vs. untreated
- 30-40 year-olds treated vs. untreated.

Then average across group

Simple Table Example:

| Age group | Avg Job Rate (Treated) | Avg Job Rate (untreated) | Difference |
|---|---|---|---|
| 20~30 | 0.80 | 0.60 | 0.20 |
| 30~40 | 0.70 | 0.60 | 0.10 |
| | | | 0.15 |

(3) Regression Adjustment

# What is Regression Adjustment?

Goal:

To estimate the causal effect of a treatment (like a new UI, a policy, a vaccine, etc) by statistically adjusting for differences between treated and untreated people.

# Why do we need adjustment?

Because in observational data, people are not randomly assigned to treatment. There may be confounders - variables that affect both:

- who get treated.
- and the outcome you care about.

# Simple Example

You want to know:

/ Did a coding bootcamp cause people to get better jobs?

Problem: people who chose the bootcamp may already be:

- younger
- more motivated
- more educated.

So you can't just compare

Avg (Job Rate / Bootcamp) - Avg (Job Rate / No Bootcamp)

This difference include bias from age, motivation, etc.

☑ Solution: Multiple Linear Regression.

Basic idea:

use a model like:

$$\text{Job Outcome} = B_0 + B_1 \cdot \text{Bootcamp} + B_2 \cdot \text{Age} + B_3 \cdot \text{Education} + \cdots$$

- $B_1$: estimates the causal effect of Bootcamp holding age and education constant.
- $B_2, B_3$: control for age and education.

# Assumption:

Regression assumes:

- All important confounders are included
- Their relationship to outcome is linear or correctly modeled
- No unobserved confounders remain.

Otherwise → your causal effect estimate will be biased.

# Final Summary Table

| Term | Meaning |
|---|---|
| Regression Adjustment | Statistically controls for confounders. |
| What it gives | Estimated causal effect of treatment. |
| Main assumption | All confounders are included and modeled properly. |
| What can go wrong | Missing confounders & wrong functional form (non linearities) |

Regression adjustment is used when:

- You have good observational data.
- You believe you can measure most key variables that affect both treatment & outcome.

## (3) Propensity Score Matching (PSM)

| # | Covariate Matching | vs | PSM |
|---|---|---|---|
| (1) What are you matching on | Raw features (age, income, etc) | | A single score = chance of being treated. |
| (2) Do people need to be exactly similar | Yes, in each feature. | | Difference in predicted probability of treatment |
| (3) How is similarity measured? | Distance between values | | Difference in predicted probability of treatment |
| (4) What's used for matching | (Age = 25, income 40k) | | Score = 0.78 (estimated from features) |

| | | |
|---|---|---|
| (5) When does it fail? | When exact matches are hard to find. | When model to estimate score is mis-specified. |
| (6) Best used when? | Few, clean variables | Many or messy variables. |
| (7) Real-life Analogy | Find someone who looks just like me | Find someone who would have made the same choice as me. |

# What is Propensity Score?

> The propensity score is the probability that a person receives ~~that~~ the treatment, given their observed characteristics.

⇒ It is the chance that someone would get the treatment, based on their background (like age, education, income).

Mathematically $e(X) = P(T = 1 \mid X)$

where,

- $T = 1$ : person got treatment.
- $X$ : observed covariates (age, income, etc).

So instead of matching on many X's, we just match treated and untreated people with similar propensity scores.

# Visual Example

| Person | Age | Education | Treated? | Propensity Score |
|---|---|---|---|---|
| A | 24 | No | ✓ Yes | 0.75 |
| B | 25 | No | ✗ No | 0.76. |
| C | 30 | Yes | ✓ Yes | 0.85 |
| D | 29 | Yes | ✗ No | 0.84 |

⇒ All 4 people have a propensity score

⇒ Only 2 were treated

⇒ You can match A ⇔ B and C ⇔ D.

# Is the Propensity Score only for Untreated People?

No.

The propensity score is computed for every person both treated and untreated.

# How Propensity Score is calculated.

Step 1: Collect Data

| Age | Education | Income | Treated | Job- |
|---|---|---|---|---|
| 23 | 0 | 30000 | 1 | 1 |
| 25 | 1 | 35000 | 0 | 0 |
| -- | -- | -- | -- | -- |

Step 2: Train a model to predict treatment

We use logistic regression to model:

$$P(\text{treated} = 1 \mid \text{features}) = \text{propensity score}.$$

That is:

> "Given this person's age, education, income, — how likely were they to be treated?".

| Age | Edu | Income | Treated | Propensity score |
|---|---|---|---|---|
| 23 | 0 | 30,000 | 1 | 0.68 |
| 25 | 1 | 35,000 | 0 | 0.23 |
| -- | -- | -- | -- | -- |

Step 3: Match based on the score.

Match each treated person to an untreated person with similar score (within a small difference)

You can use:
- Nearest neighbor
- Caliper matching
- Stratified matching.

| # Concept | PSM | vs | Regression Adjustment |
|---|---|---|---|
| Tool | Matching people | | Building a formula |
| Goal | Match similar people, compare their outcomes | | Use math to adjust outcomes. |
| What it does | Matches treated and untreated people with similar background. | | Builds a model to predict outcomes using treatment + background. |
| Think of it like | finding your "twin" | | Using a formula to remove background effects. |

# Role of Regression Adjustment for both covariant matching and PSM.

⇒ Quick Answer:

> Regression adjustment is a separate method from matching. But it can be used alongside both.

- ☑ Covariate Matching and,
- ☑ Propensity score Matching (PSM) to improve accuracy.

→ Think of it this way:

| Tool | what it does |
|---|---|
| Matching (covariates or PSM) | Tries to balance groups first by making them look similar. |
| Regression Adjustment | Then removes any leftover bias using a formula. |

# Confounding.

→ Confounding happens when something affects both treatment and the outcome, make it look like the treatment caused the outcome - when it really didn't (or not fully).

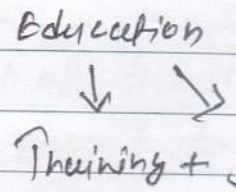
Example: Job Training Programs

| Variable | what it does. |
|---|---|
| Treatment: Training | People choose ~~to~~ take it. |
| Outcome: Gota job | what we want to measure. |
| Confounder: Education level | Affect both: educated people may join training more, and get jobs more. |

So:
- If you compare trained vs untrained without adjusting
- you'll wrongly give training credit for what education did.

That is confounding

# Visual: Confounding Graph.




- Education causes both training & job.
- That is why training and job look related, even if training has no real effect.

# What Matching (covariate or PSM) Does?
It removes confounding by balancing the confounder.

So you:

- Match a treated person with a similar untreated person (same education, same age etc).
- Now the only difference is one got training, one didn't.

That is why matching helps us estimate true causal effects.

=) A confounder is a variable that influence both the treatment and the outcome, and if not adjusted for, it can mislead you about what caused what.

## (4) Difference-in-Differences (DiD)

# What are we trying to do?

You want to know:

1? Did something cause a change - like a new law, a new feature, a training, etc?

But things change over time anyway, right?

So we need a method to separate the effect of the treatment from natural time changes.

That method is called

☑ Difference-in-Differences (DiD).

# Real-life Example (No math first)

Let's say you want to know

" Did giving free lunches in school improve test scores? ".

You do this:

- choose two similar schools.
- School A gets free lunch
- School B does not.

You measure scores:

| Time | School A (Free Lunch) | School B (No Lunch) |
|---|---|---|
| Before | 60 | 55 |
| After | 83 | 65 |

Now pause.

You see:

- School A improved from 60 → 85 (+25)
- School B improved from 55 → 65 (+10)

so, part of the increase may just be normal (maybe both schools improved anyway)

So what is real effect of free lunch?

Subtract the normal trend:

DiD Estimate = (85 − 60) − (65 − 55) =

25 − 10 = 15

☑ That is points is the estimated causal effect of the free lunch.

# Why we do this

we use DiD to remove the shared growth or decline that's happening over time to everyone

without DiD:

- You might wrongly credit the treatment (free lunch) for natural improvements.

# Summary Table.

| Concept | Meaning. |
|---|---|
| Why use DiD | → To isolate true treatment effect from natural trends. |
| What it compares! | → change in treated vs. change in untreated. |
| Key assumptions | → Parallel trends: both groups were changing similarly before. |
| Output | → Estimated treatment effect. |

(5) Concept links.

- Unconfoundedness (from prior topic) is the foundation for all these designs.
- Counterfactuals: You're trying to estimate the outcome a person didn't get.
- Causal inference – using these designs to approximate counterfactuals.

(6) Final Summary Tables

| Design | Key idea | Good for |
|---|---|---|
| Matching | Find similar untreated for each treated | Low-dimensional covariates |
| Stratification | Compare within groups | Categorical/binned variables. |
| Regression | Predict outcome adjusting for covariates. | Continuous control + linearity. |
| Propensity score | Match using treatment probability | High-dimensional covariates. |
| Difference-in-Difference | Use before/after comparisons | Time series panel data. |

| | | |
|---|---|---|
| 4 | 1 | 0 |
| 5 | 0 | 0 |
| 6 | 1 | 1 |
| 7 | 0 | 0 |
| 8 | 1 | 1 |
| 9 | 0 | 1 |
| 10 | 0 | 0 |

Step 2: Confusion Matrix

| | Rater 2 = 1 | Rater 2 = 0 | Total |
|---|---|---|---|
| R1 = 1 | 4 | 1 | 5 |
| R1 = 0 | 1 | 4 | 5 |
| Total | 5 | 5 | 10 |

Step 3: Calculate Cohen's κ

$$\kappa = \frac{P_o - P_e}{1 - P_e}$$

(a) Observed agreement $P_o$

$$P_o = \frac{\text{agree cases}}{\text{total}} = \frac{4+4}{10} = 0.8$$

(They agreed on 4 "good" and 4 "bad" → 8/10 total agreement)

(b) Expected agreement $P_e$

Based on independent probabilities of ratings:

- Probability both say "1":

$$P_{11} = P(R1=1) \cdot P(R2=1) = \frac{5}{10} \cdot \frac{5}{10} = 0.25$$

- Probability both say "0":

$$P_{00} = \frac{5}{10} \cdot \frac{5}{10} = 0.25$$

so:

$P_e = 0.25 + 0.25 = 0.5$

(c) Final K:

$$K = \frac{0.8 - 0.5}{1 - 0.5} = \frac{0.3}{0.5} = 0.6.$$

☑ Interpretation

- $K = 0.6$ = "Moderate to Substantial Agreement"
- Much better than chance
- This means the human evaluation is statistically trustworthy.

⇒ Real World Use

If two UX designers or raters label app screens for "clean layout", and you get $K > 0.6$, you can trust their judgement, and maybe use the labels for:

- Training ML models
- Validating UI changes in A/B testing
- Feeding into a composite score with logs-based data.

# Fleiss Kappa:

It measures how consistently multiple raters assign categorical labels (e.g. "good" / "bad") across many items - correcting for agreement by chance.

Example setup (3 Raters, 5 items)

Each item is rated as either:

- 1 = ☑ Good
- 0 = ☒ Bad.

Step 1: Manual Math Breakdown

[V] Goal => You have multiple raters (e.g. 3) labeling each item, and you want to compute:
How much do they agree beyond chance?
We'll walk through the calculation for item 1, and then generalize to Fleiss' Kappa.

1.1 Given: item 1 Rating

| Rating Label | Count |
|---|---|
| Bad (0) | 1 |
| Good (1) | 2 |

So 3 raters: 1 said "Bad", 2 said "Good".

From,

| item | # of raters said Good | # of raters said Bad. |
|---|---|---|
| 1 | 2 | 1 |
| 2 | 3 | 0 |
| 3 | 0 | 3 |
| 4 | 1 | 2 |
| 5 | 2 | 1 |

1.2: Count Number of Rater Pairs
If you have 3 raters judging the same items, then you can form pairs of raters to see whether each pair agrees or disagrees.
We're not counting individual ratings - we're checking pairwise agreement.

- Let's label the 3 Raters A, B, C
- How many pairs of raters can you make?
  You can pair:
  - A with B
  - B with C
  - A with C

so there are 3 unique pairs total.

that's what this means mathematically:

$$\binom{3}{2} = \frac{3!}{2!(3-2)!} = \frac{3 \cdot 2 \cdot 1}{2 \cdot 1 \cdot 1} = 3$$

☑ so for 3 raters, we have 3 pairs to check agreement on.

# why do we need pairs?

Because agreement happens between pairs

- if 2 raters both say "Good", that's an agreement.
- if 1 says "Good" and the other "Bad", that's disagreement.

so we ask:

| "out of all 3 possible pairs, how many agreed?"

That's why we calculate total pairs first.

Step 2 Count How Many Pairs Agree.

we count how many rater pairs gave the same rating.

- 2 raters said "Good" → there is 1 agreeing pair among them

$$\binom{2}{2} = 1.$$

- 1 rater said "Bad" → can not form a pair

$$\binom{1}{2} = 0$$

so

Agreeing Pairs = 1 + 0 = 1.

Step 3: proportional of Agreement (for item 1)

$$P_1 = \frac{\text{Agreeing Pairs}}{\text{Total Pairs}} = 4/3$$

Step 4: General formula for any item

if an item has n = 3 raters, and the counts for each category (e.g. good, bad, fair, etc) are $n_1, n_2 \ldots n_k$, then:

$$P_i = \frac{1}{n(n-1)} \sum_{j=1}^{k} n_{ij}(n_{ij}-1)$$

where:

- $P_i$ : Agreement for item i
- n : Number of raters per item
- k : Number of categories
- $n_{ij}$ : Number of raters who choose category j for item i

This counts how many pairs agree on the same category.

Solutions

we had:

- category 0 (Bad) : $n_0 = 1$
- Category 1 (Good) : $n_1 = 2$

so!

$$P_1 = \frac{1}{3 \cdot 2}[1 \cdot (1-1) + 2 \cdot (2-1)] = 1/6 (0+2) = 2/6$$

$$= 1/3$$

✓ same result.

Steps: Interpreting k

| Kappa | Interpretation |
|---|---|
| < 0.0 | Poor agreement |
| 0.0 - 0.20 | Slight agreement |
| 0.21 - 0.40 | Fair agreement |
| 0.41 - 0.60 | Moderate agreement |
| 0.61 - 0.80 | Substantial agreement |
| 0.81 - 1.00 | Almost perfect agreement |

(4) Design checklist (math meets practice)

This is a fully-worked mathematical example that shows how Human Evaluation is turned into a statistically rigorous signal.

1. Sampling items

Assume we randomly select 5 search queries (items) from a long tail so results are representative.

| item(query) | $id_1$ | $id_2$ | $id_3$ | $id_4$ | $id_5$ |
|---|---|---|---|---|---|
| K2 | 1 | 2 | 3 | 4 | 5 |

No math yet - this is design.

* Long tail means rare search queries, niche products, infrequent user actions ⇒ all the uncommon queries combined as the head means most common (e.g. 'weather', 'facebook').

## 2. Number of Rater per item (R=3)

# The sentence in Question:

> "Three trained judges rate control vs. treatment side-by-side on a 1-5 quality scale ... Hence, doubling raters halves the variance.

# What it means in plain English

You have:

- 5 items (e.g. search queries or UI screens)
- Each item is shown to 3 raters
- Each rater gives a score (from 1 to 5) for both: ↳ Control version (the old version).
  ↳ Treatment version (the new version)

So for each item, you get 3 scores for control and 3 scores for treatment. You then take the average score of the 3 raters for each ~~item~~ items.

Now the key point:

> The more raters you have, the more stable and less noisy the average score becomes.

That is what they mean what they say:

* * > "Doubling raters halves the variance". * *

# Example: Variance Shrinking by More Raters.

Let's say you're rating a movie and 1 person says:

- "It's 5 stars!"

But that person might be biased or random.

Now suppose 3 people rate:

- 5, 4, 4 → Average ≈ 4.33

That's a more reliable estimate.

If 6 people rate it, it becomes even more stable
e.g:

- 5, 4, 4, 5, 4, 4 → Average = 4.33 again with less random fluctuation.

# Math Behind it (optional, but here's what that sentence is saying):

Let's say:

- $x_1, x_2, x_3$ are the 3 ratings for an item's control version.
- The mean score is:

$$\bar{x} = \frac{x_1 + x_2 + x_3}{3}$$

- The variance of the mean is:

$$Var(\bar{x}) = \frac{\sigma^2}{12}$$

where:

- $\sigma^2$ = the natural variability between raters
- R = number of raters.

# So the logic is:

- If R = 3 → variance = $\sigma^2/3$
- If R = 6 → Variance = $\sigma^2/6$

[V] Doubling the raters reduces the noise in the average by half.

# Deriving

$$Var(\bar{X}) = \frac{\sigma^2}{R}$$

Step 1: what is variance?

For any random variable $X$:

$$Var(X) = E[(X-\mu)^2]$$

It measures how spread out the values of $X$ are around the mean $\mu$.

Step 2: Suppose you have $R$ independent random variables

Let's say you collect $R$ independent ratings $X_1, X_2, \ldots X_R$, each with:

- the same mean $\mu$
- and same variance $\sigma^2$

Now compute their average:

$$\bar{X} = \frac{1}{R}\sum_{i=1}^{R} X_i$$

What is the variance of this mean?

Step 3: Use a key property of variance

For independent random variables

$$Var\left(\sum X_i\right) = \sum Var(X_i)$$

So:

$$Var(\bar{X}) = Var\left(\frac{1}{R}\sum_{i=1}^{R} X_i\right) = \frac{1}{R^2}\sum_{i=1}^{R} Var(X_i)$$

$$= \frac{1}{R^2} \cdot R \cdot \sigma^2 = \frac{\sigma^2}{R}$$

Step 4: Final Result

$$Var(\bar{X}) = \frac{\sigma^2}{R}$$

✓ This means the more ratings ($R$) you use, the less noisy your average rating becomes.

## 8. Blocking & Randomization (Paired Design)

# Goal

To test whether treatment is better than control, based on human rating. You have 5 items (e.g. 5 search queries), and 3 raters rate both control and treatment for each item on a scale of 1 to 5.

# KEY IDEA: "Paired Design"

Instead of comparing all control scores vs. all treatment scores (like unpaired t-test), we compare:

> Per-item difference in mean rating between treatment and control.

This gives us one clean number per item, and filters out the noise caused by item difficulty.

# Step-by-step walkthrough with Real Data

Data: Rating by 3 people for 5 items

| item | Control Ratings | Treatment Ratings |
|---|---|---|
| 1 | 4, 4, 3 | 5, 5, 4 |
| 2 | 2, 3, 2 | 3, 4, 3 |
| 3 | 5, 4, 5 | 5, 3, 5 |
| 4 | 3, 3, 4 | 4, 4, 5 |
| 5 | 4, 4, 3 | 4, 5, 4 |

Step 1: Mean rating per item

Control means:

$$\bar{x}(C) = \left( \frac{4+4+3}{3}, \frac{2+3+2}{3}, \frac{5+4+5}{3}, \frac{3+3+4}{3}, \frac{4+4+3}{3} \right)$$

$$= (3.67, 2.33, 4.67, 3.33, 3.67).$$

Treatment means:

$$\bar{X}^{(T)} = \left( \frac{5+5+4}{3}, \frac{3+4+3}{3}, \frac{5+5+5}{3}, \frac{4+4+5}{3}, \frac{4+5+4}{3} \right) = (4.67, 3.33, 5.00, 4.33, 4.33)$$

Step 2: Compute differences for each time

$$D_i = X_i^{(T)} - X_i^{(C)}$$

So,

$$D = (1.00, 1.00, 0.33, 1.00, 0.67)$$

Step 3: Compute mean and standard deviation of differences

$$\bar{D} = \frac{1+1+0.33+1+0.67}{5} = 0.80$$

Now compute sample standard deviation $S_D$:

- First compute squared deviations from the mean:

$(1-0.8)^2 = 0.04$, ..... $(0.67-0.8)^2 = 0.02$

Sum $= 0.04 + 0.04 + 0.22 + 0.04 + 0.02 = 0.36$

Divide by degrees of freedom $n-1 = 5-1 = 4$

$$S_D = \sqrt{\frac{0.36}{4}} = \sqrt{0.09} = 0.30$$

Step 4: Compute Standard error (SE)

$$SE = \frac{S_D}{\sqrt{n}} = \frac{0.30}{\sqrt{5}} = \frac{0.30}{2.236} \approx 0.134.$$

Step 5: Compute t-statistics

$$t = \frac{\bar{D}}{SE} = \frac{0.80}{0.134} \approx 5.97$$

Compare with t-critical ($df = 4$): $t_{0.05}, df = 4$

$\approx 2.776$

☑ Since $5.97 > 2.776$ we reject the null hypothesis → treatment is significantly better than control.

# Why Paired Design is Better?

If you had just thrown all 15 control and 15 treatment ratings together, and done a regular t-test:

- Would get $t \approx 2.66$ - just barely significant
- Because it wouldn't cancel out item difficulty noise

But the paired design subtracts noise at the item level, making the test sharper and more trustworthy

## Summary

| Test type | SE | t-value | Conclusion |
|---|---|---|---|
| Unpaired | 0.30 | 2.65 | borderline significant |
| paired ☑ | 0.13 | 5.97 | Strongly significant |

## 4. Gold Questions (Spam Detection)

# What are "Gold Questions"?

- These are secret test items mixed into the rating task.
- They have a known correct answer, usually agreed on by experts.
- Used to detect unreliable raters (e.g. spammers/careless workers).

Think of them like answer key questions in a exam that help catch cheaters.

# Example Setup

Let's say:

- You give a rater 20 total rating tasks.
- 2 of those are "gold" questions, where the true answer is known (e.g. score = 5)
- If a rater gives the gold questions poor ratings (e.g. 1, 2, 3), they might be guessing or spamming

# Use Binomial Test

We want to statistically test:

> "Is this rater worse than what we'd expect by random chance?"

⇒ Hypothesis setup

- Each gold question is like a coin flip, with some chance $p$ of being correct.
- Let's assume a spammer guesses randomly and has 30% chance to answer each gold question correctly → $p = 0.3$

If the rater got 0 out of 2 gold items correct, is that too bad to be chance?

⇒ Use a Binomial Tail Test

☑ Step-by-step

We model this as a Binomial Distribution:

$$X \sim \text{Binomial}(n = 2, p = 0.3)$$

This means:

- The rater answers 2 question ($n = 2$)
- Each answer has a 30% chance ($p = 0.3$) of being correct.
- $X$ is the number of correct answers (can be 0, 1, 2).

Formula:

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

Case 1: Getting 0 correct ($X = 0$)

$$P(X = 0) = \binom{2}{0} \cdot (0.3)^0 \cdot (1-0.3)^2 = 1 \cdot 1 \cdot (0.7)^2$$

$$= 0.49$$

case 2 : Getting 1 correct ($X=1$)

$$P(X=1) = \binom{2}{1} \cdot (0.3)^1 \cdot (0.7)^1 = 2 \cdot 0.3 \cdot 0.7$$

$$= 0.42$$

case 3 : Getting 2 correct ($X=2$)

$$P(X=2) = \binom{2}{2} \cdot (0.3)^2 \cdot (0.7)^0 = 1 \cdot 0.09 \cdot 1 = 0.09$$

Interpretation

if a rater :

- Gets 0 or 1 correct → could easily happen by random guessing (42% chance)
- Gets 2 correct → very unlikely to happen by chance (only 9%)

⇒ so, we trust raters who get 2/2 gold questions right.

# Final Rule

> If a rater gets too many gold answers wrong, and the probability they could fail that badly just by chance is small, we reject their data.

## 5. Aggregate Score (Median to resist outliers)

let's say 3 human raters give content-relevance ratings for a search result :

| Rater 1 | Rater 2 | Rater 3 |
|---|---|---|
| 4 | 4 | 1 ← error/spam |

# If you use the mean

$$\text{mean} = \frac{4+4+1}{3} = \frac{9}{3} = 3.0$$

that 1 rating pulled the average very down (from near 4 to 3)

# If you use the median

Sorted value : 1, 4, 4

→ Median = 4

The outliers has no affect on the median unless more than 50 % of ratings are bad.

# What is "a breakdown point" ?

The breakdown point of a statistics is how much bad data it can tolerate before it becomes useless.

- Mean has a breakdown point of 0% : One extreme value can ruin it.
- Median has a breakdown point of 50% : up to half the data can be corrupted and the results still robust.

(5) Where it overlaps other techniques.

| Overlap partner | Synergy |
|---|---|
| Logs-based analysis | Logs show CTR ↑; human eval confirms relevance ↑ (or flags click-bait) |
| User-Experience Research (UER) | UER discovers why raters marked items low; feeds better rating rubrics. |
| Surveys | Post-experiment survey answers can be compared against expert rating to check external validity |

(6) One-sentence takeaway.

Human evaluation injects subjective but essential quality signals into an experimentation pipeline; with solid sampling and reliability stat (e.g. Cohen's κ), these signals become rigorous enough to guide launch decisions.

(1) What is User Experiance Research (UER)?

→ UER is a qualitative research method that focus on how users feel and behave when interacting with a product or feature. It complements A/B testing, which is quantitative.

# Why it's needed:

A/B tests can tell you what changed (e.g. click-through rate went up), but UER helps you understand why it changed — or how users experienced it.

# Simple Explanation

Think of A/B testing as watching security camera footage
| You see users walk into a store and whether they buy or not - what happened. ~~UER is like interviewing the shopper after.~~

UER is like interviewing the shopper after:
| "How was your experience? Did you find the layout confusing? were the signs clear?

# Where UER helps (examples):

~~Think of A/B testing as watching security camera footage~~:

| A/B Testing Tells You | UER Can Reveal |
|---|---|
| Users click less | They felt overwhelmed or confused. |
| Users bounce more | They didn't understand the new layout |
| Time on page increased | They were lost, not engaged. |

New feature didn't help  User didn't even notice it.

(2) Math concept related to UBR

| Math Concept | How it Connects to UBR |
|---|---|
| Latent Variables | → Hidden emotional states affects behavior. |
| Simpson's Paradox | → Averages hide subgroup behavior |
| Causal Inference | → UBR informs mechanism of treatment effects. |
| Bayesian Thinking | → Prior beliefs from guide interpretation of metrics. |
| Variance Decomposition | → UBR helps understand if changes affect all or just some users. |
| Information Theory | → UBR detects if users received the UI signal. |

The most foundational math concept applied in USER EXPERIENCE RESEARCH is Latent Variables.

# Latent Variable why?

→ UBR is fundamentally about uncovering unobserved causes — like user confusion, emotional reactions, attention level — that influence observed behavior such as:

- clicks
- Time on page
- Task success
- Scroll depth.

Since these hidden factor aren't directly logged, they are latent and must be inferred.

Unit 11

## (5) Pitfalls in observational causal studies.

This section outlines the dangers and limitations when trying to estimate causal effect using observational data, especially when controlled experiments (A/B tests) aren't possible.

### 1. Confounding Variables

- Problem: Unmeasured variables may affect both treatment and outcomes.
- Example: Highly motivated users might be more likely to adopt a new feature and perform better — so it's not the feature causing better outcomes.

☑ What to do: Adjust using methods like:

- Propensity Score Matching
- Regressions

### 2. Selection Bias

- Problem: When users self-select into treatment/control, your group may differ in important unobserved ways.
- Example:

> users who upgrade to a paid plan may already be more active and loyal. Comparing them with free users can mislead you.

☑ Fixes:

- Try to replicate randomization via matching
- Use techniques like inverse probability weighted.

## (3) Model Misspecification

- Problem: If your regression or scoring model assumes the wrong relationship (e.g. linear when it's not), estimates will be off.

  Bad model = bad causal claims.

☑ What helps:
- Use flexible model (e.g. splines, trees)
- Validate predictions, check residuals.

## (4) Over controlling (Adjusting for Mediators or Colliders)

- Problem: including variables on the causal path or common effects introduces bias.

  Don't adjust for
- Mediators (e.g. feature → engagement → revenue)
- Collider (e.g. bug causes drop and fix leads to bounce → bounce is a collider)

☑ Use causal diagrams to decide what not to control.

## (5) Simpson's Paradox

- Problem: The treatment looks beneficial in the aggregate but harmful in subgroups — or vice versa.

- Example: A feature seems positive overall but is actually hurting mobile users while helping desktop ones.

☑ Always stratify or segment results by meaningful user groups.

(6) No Ground Truth.

- Observational studies rarely have "truth" to compare against,
- so even a fancy causal estimate can't be verified,

☑ if possible, validate with.

- small randomized trials.
- Back-testing on historical rollouts.

## (4) Sidebar: Refuted Observational Causal Studies.

### (1) What's This Section About?

This sidebar highlights a big warning:

⚠ Many famous observational studies (non-experiments) made bold causal claims...
→ But when tested later with randomized experiment (A/B test or RCTs) — most were proven wrong.

(1) Hormone Replacement Therapy (HRT)

# Real Examples (from the book and cited studies).

- Observational studies: HRT reduces heart disease risk.
- RCT (Women's Health Initiative) found: HRT increased heart attacks, strokes, and breast cancer.
- Why wrong? Healthier women were more likely to get HRT → confounding bias.

(2) Anti-oxidants Prevent Cancer

- observational studies said: Beta-carotene lowers cancers risk.
- RCT, showed: No benefit, and even higher cancer rates in smokers who took it.
- Why wrong? People who take vitamins also tend to live healthier in other ways → confounders again.

### (2) Key Insights.

Observational data can fool you because it doesn't control for hidden variables (confounders), self-selection, or natural trends. Randomized experiments, on the other hand, neutralize these issues.

So... why do observational studies fail?
① confounding ② selection bias ③ Overfitting or model bias ④ Reverse causality.

③ Real-world Analogy
Let's say
"Kids who bring apples to school score higher on math test"
But really: Smart, organized families are more likely to pack apples and help with homework.

☑ The apple isn't magic — it's a proxy for the real cause.

④ Related Topics You've learned.

| Concept | How it related here. |
| --- | --- |
| Confounding bias | Main reasons observational studies go wrong. |
| Matching / PSM / DiD | Attempts to fix observational limitations. |
| Internal Vs. External validity | why results often don't replicate |
| Importance of A/B testing | To confirm observation claims. |